

Dirac Delta Function

 $\delta(x)$

Quick revision note • definition, properties, proofs, worked examples

1. What it is

An idealised **infinitely tall, infinitely thin spike** with unit area, used for a point charge, impulsive force, point mass or short-range force. It is not a true function (“improper”); it only makes sense *inside an integral*. It is the continuous version of the Kronecker delta δ_{ij} .

Definition (1D)

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1$$

Shifted:

$$\delta(x - a) = \begin{cases} 0, & x \neq a \\ \infty, & x = a \end{cases} \quad \int_{-\infty}^{\infty} \delta(x - a) dx = 1$$

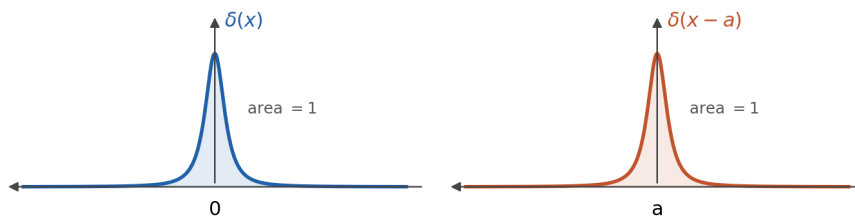
Definition (3D)

$$\delta(\vec{r} - \vec{r}_0) = \begin{cases} 0, & \vec{r} \neq \vec{r}_0 \\ \infty, & \vec{r} = \vec{r}_0 \end{cases}$$

$$\int_{\text{all space}} \delta(\vec{r} - \vec{r}_0) dv = 1$$

$$\text{Over a volume } V: \int_V \delta(\vec{r} - \vec{r}_0) dv = \begin{cases} 1, & \vec{r}_0 \in V \\ 0, & \vec{r}_0 \notin V \end{cases}$$

In Cartesian form: $\delta(\vec{r}) = \delta(x) \delta(y) \delta(z)$.



2. Sifting property

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a) \quad (\text{and } f(0) \text{ when } a = 0)$$

Rule: δ picks out f at the point where its argument is 0, provided that point lies inside the limits; otherwise the integral is 0.

3. Other ways to write δ

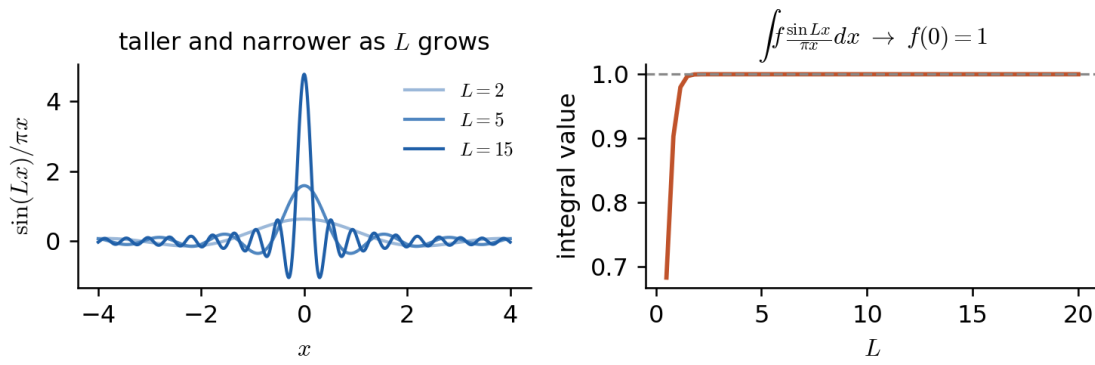
Integral form

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk, \quad \delta(x - a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-a)} dk, \quad \delta(\vec{r}) = \left(\frac{1}{2\pi}\right)^3 \int e^{i\vec{k} \cdot \vec{r}} d^3k$$

Limit form $\delta(x) = \lim_{L \rightarrow \infty} \frac{\sin Lx}{\pi x}$

Proof. Cut the integral form off at $\pm L$:

$$\delta(x) = \frac{1}{2\pi} \lim_{L \rightarrow \infty} \int_{-L}^L e^{ikx} dk = \frac{1}{2\pi} \lim_{L \rightarrow \infty} \left[\frac{e^{ikx}}{ix} \right]_{-L}^L = \frac{1}{2\pi} \lim_{L \rightarrow \infty} \frac{2i \sin Lx}{ix} = \lim_{L \rightarrow \infty} \frac{\sin Lx}{\pi x}$$



Left: $\sin(Lx)/\pi x$ for growing L (area stays 1). Right: numerical check that $\int_{-\infty}^{\infty} f \frac{\sin Lx}{\pi x} dx \rightarrow f(0)$, using the test function $f = e^{-x^2/8}(1+x/2)$ with $f(0) = 1$.

4. Properties with proofs

(a) Even: $\delta(-x) = \delta(x)$

Let $y = -x$:

$$\int_{-\infty}^{\infty} f(x) \delta(-x) dx = \int_{-\infty}^{\infty} f(-y) \delta(y) dy = f(0) = \int_{-\infty}^{\infty} f(x) \delta(x) dx \Rightarrow \delta(-x) = \delta(x).$$

(b) Scaling: $\delta(ax) = \frac{\delta(x)}{|a|}$ **(3D:** $\delta(a\vec{r}) = \frac{\delta(\vec{r})}{|a|^3}$)

Put $y = ax$. For $a > 0$:

$$\int_{-\infty}^{\infty} f(x) \delta(ax) dx = \frac{1}{a} \int_{-\infty}^{\infty} f\left(\frac{y}{a}\right) \delta(y) dy = \frac{f(0)}{a}.$$

For $a = -b < 0$: $y = -bx$, $dx = -dy/b$, and the limits swap, cancelling the minus sign:

$$\int_{-\infty}^{\infty} f(x) \delta(ax) dx = \frac{1}{b} \int_{-\infty}^{\infty} f\left(-\frac{y}{b}\right) \delta(y) dy = \frac{f(0)}{b} = \frac{f(0)}{|a|}.$$

In both cases the result is $f(0)/|a| = \int_{-\infty}^{\infty} f(x) \frac{\delta(x)}{|a|} dx$.

(c) Derivative: $\int_{-\infty}^{\infty} f \delta' dx = -f'(0) \Rightarrow x \delta'(x) = -\delta(x)$

By parts (boundary term vanishes because $\delta = 0$ at $\pm\infty$):

$$\int_{-\infty}^{\infty} f(x) \delta'(x) dx = [f \delta]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x) \delta(x) dx = -f'(0).$$

Now take the test function $x f(x)$ in place of f : $\frac{d}{dx}(xf) \Big|_0 = f(0)$, so

$$\int_{-\infty}^{\infty} f(x) [x \delta'(x)] dx = -\frac{d}{dx}(xf) \Big|_{x=0} = -f(0) = \int_{-\infty}^{\infty} f(x) [-\delta(x)] dx \Rightarrow x \delta'(x) = -\delta(x).$$

5. Key result in 3D: $\nabla \cdot \frac{\vec{r}}{r^3} = 4\pi \delta(\vec{r})$

Away from the origin ($\vec{r} \neq 0$):

$$\nabla \cdot \frac{\vec{r}}{r^3} = \frac{\nabla \cdot \vec{r}}{r^3} + \vec{r} \cdot \nabla \frac{1}{r^3} = \frac{3}{r^3} - \frac{3}{r^3} = 0.$$

At the origin it is indeterminate, so apply the divergence theorem to a small sphere of radius R :

$$\int_V \nabla \cdot \frac{\vec{r}}{r^3} dv = \oint \frac{\vec{r}}{r^3} \cdot d\vec{s} = \frac{1}{R^2} \oint ds = \frac{4\pi R^2}{R^2} = 4\pi.$$

This holds *however small* R is: the divergence is zero everywhere except at the origin, yet its total is 4π . So we may replace it by $4\pi \delta(\vec{r})$.

$$\nabla \cdot \frac{\vec{r}}{r^3} = 4\pi \delta(\vec{r}), \quad \nabla^2\left(\frac{1}{r}\right) = -4\pi \delta(\vec{r}), \quad \delta(\vec{r}) = \frac{1}{4\pi} \nabla \cdot \frac{\vec{r}}{r^3} = -\frac{1}{4\pi} \nabla^2 \frac{1}{r}$$

6. Quick reference

$\int_{-\infty}^{\infty} f(x) \delta(x-a) dx = f(a)$	sifting	$\delta(-x) = \delta(x)$	even
$\delta(ax) = \delta(x)/ a $	scaling	$x \delta'(x) = -\delta(x)$	derivative identity
$\int_{-\infty}^{\infty} f \delta' dx = -f'(0)$	derivative sifting	$\nabla \cdot (\vec{r}/r^3) = 4\pi \delta(\vec{r})$	3D point source

7. Worked examples

- 1(i) $\int_{-\infty}^{\infty} x^3 \delta(x-2) dx = 2^3 = 8$ centre $x = 2$ inside
- 1(ii) $\int_0^1 x^3 \delta(x-2) dx = 0$ centre $x = 2$ outside $[0, 1]$
- 2 $\int_{-\infty}^{\infty} e^x \delta'(x) dx = -\frac{d}{dx} e^x \Big|_{x=0} = -1$ uses $\int f \delta' = -f'(0)$
- 3 $\int_{-2}^2 (3x+2) \delta(4x) dx = \frac{1}{4} \int_{-2}^2 (3x+2) \delta(x) dx = \frac{1}{4}(2) = \frac{1}{2}$ scaling, $|a| = 4$
- 4 $\int_V |\vec{r} - \vec{b}|^2 \delta(5\vec{r}) dv = \frac{1}{125} \int_V |\vec{r} - \vec{b}|^2 \delta(\vec{r}) dv = \frac{b^2}{125}$ 3D scaling 5^3 ; origin $\in V$
- 5 $\int_0^{\pi} \cos x \delta(x - \pi) dx = \cos \pi = -1$ endpoint delta: counted as inside (see note)
- 6 $\int_{-2}^2 x^3 \delta(x+1) dx = (-1)^3 = -1$ centre $x = -1$ inside
- 7 $\int_{-2}^2 (2x+3) \delta(3x) dx = \frac{1}{3} \int_{-2}^2 (2x+3) \delta(x) dx = \frac{1}{3}(3) = 1$ centre $x = 0$ inside
- 8 $\int_0^2 (x^3 + 3x + 2) \delta(x-3) dx = 0$ centre $x = 3$ outside $[0, 2]$

Note on example 5. The delta sits exactly on the upper limit $x = \pi$. Strictly this is ambiguous: a symmetric (half-weight) convention would give $-\frac{1}{2}$. The result -1 follows the convention used in the lecture notes, where a delta at a limit counts fully. Avoid such endpoint cases in practice.